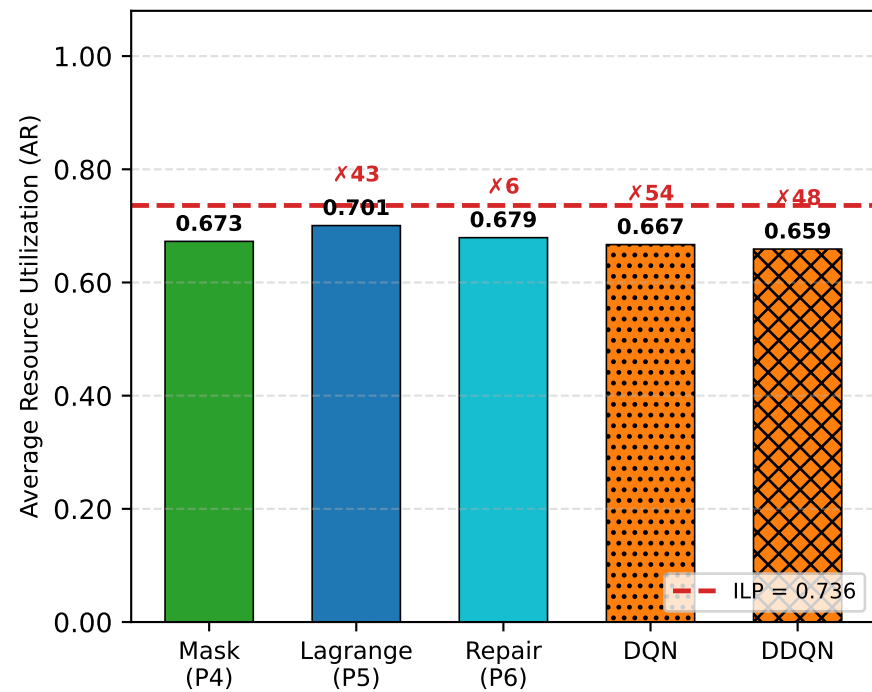


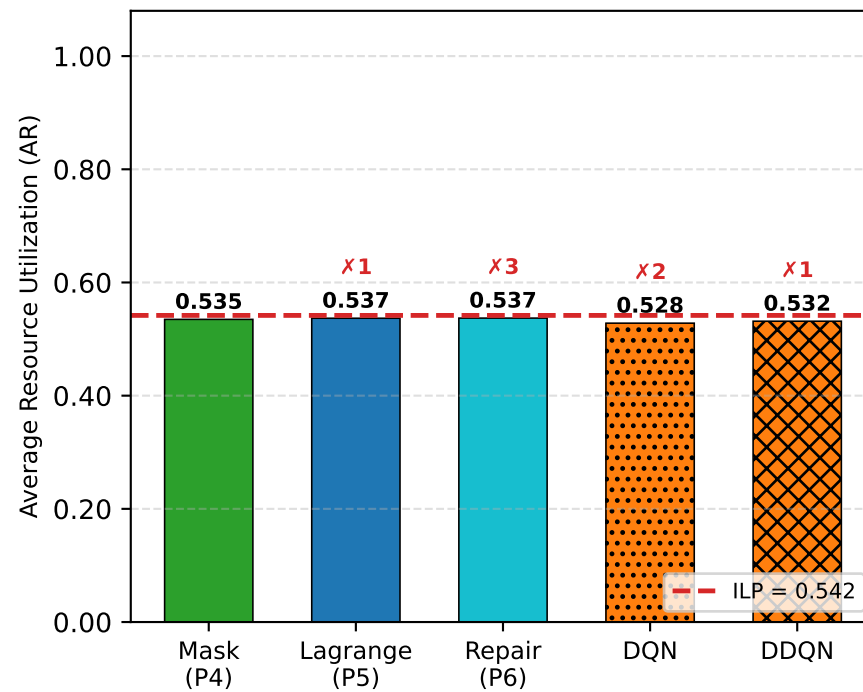
AR Comparison — Constrained Methods vs ILP Optimum

Violation count annotated above bar ($\times N$ = cap + conflict violations over 40 test episodes)

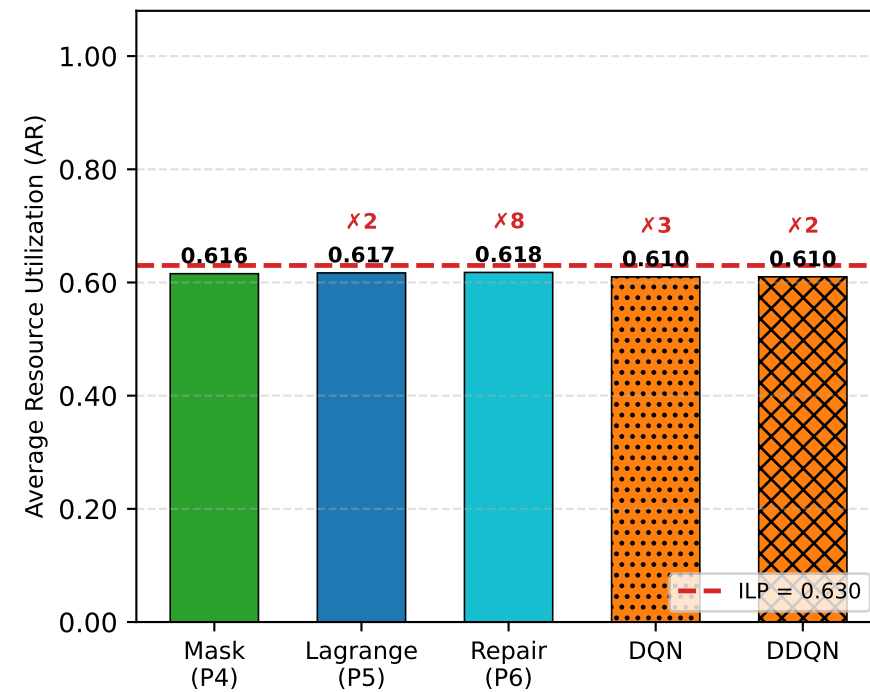
LT (N=10 ECUs < M=15 SVCs)
Resource-scarce



EQ (N=M=10)
Balanced



GT (N=15 ECUs > M=10 SVCs)
Resource-abundant



Legend: P4 — Hard mask (zero violations guaranteed) | P5 — Lagrangian / soft constraint | P6 — Repair heuristic | DQN | DDQN